

Unknown angles at a point and on a line

Four facts about angles that meet — and the equations they turn into.

LESSON 81–83

3 × 40 MIN

MATEMATIKA 6, §16

S7 5.1–5.2

LESSON CLOCK

25'

30'

35'

25'

5

The four facts	25 min
One unknown	30 min
Equations with letters	35 min
Writing reasons	25 min
Homework	5 min

LEARNING OBJECTIVES

- Use the fact that angles on a straight line add to 180° .
- Use the fact that angles at a point add to 360° .
- Use vertically opposite angles.
- Write and solve an equation for an unknown angle.

TEXTBOOK

National	pp. 227–236
Cambridge	Stage 7, pp. 55–62
Workbook	Exercise 5.2

01

Explanation

The four facts

SITUATION	FACT	REASON TO QUOTE
two angles making a straight line	they add to 180°	angles on a line
angles filling a turn	they add to 360°	angles at a point
two lines crossing	opposite angles are equal	vertically opposite
two angles making a right angle	they add to 90°	complementary angles

EVERY ANSWER CARRIES A REASON

In geometry the number alone is half the answer. “ $x = 55^\circ$, angles on a line” is the whole one, and the reason is where the mark is.

One unknown

GIVEN	FACT	WORKING	ANSWER
125° on a line with x	180°	$180 - 125$	55°
140° , 90° and x at a point	360°	$360 - 230$	130°
x vertically opposite 68°	equal	—	68°
x and 37° make a right angle	90°	$90 - 37$	53°

CHOOSE THE FACT BY THE PICTURE, NOT BY THE NUMBERS

Angles that look as if they fill a turn add to 360° even if there are five of them; angles along a line add to 180° however many there are.

Equations with letters

SITUATION	EQUATION	SOLUTION	THE ANGLES
x and $2x$ on a line	$3x = 180$	$x = 60$	60° , 120°
x , $x + 20$ and $2x$ at a point with 90°	$4x + 110 = 360$	$x = 62.5$	62.5° , 82.5° , 125°
$3x$ and $x + 40$ vertically opposite	$3x = x + 40$	$x = 20$	60°
x , $2x$, $3x$ on a line	$6x = 180$	$x = 30$	30° , 60° , 90°

GEOMETRY BECOMES ALGEBRA, AND BACK AGAIN

Write the equation from the picture, solve it with the methods of lesson 16, then put the value back into every expression to find the actual angles.

Writing reasons

STATEMENT	REASON
$x + 125 = 180$	angles on a straight line
$x = 55^\circ$	solving
$y = 68^\circ$	vertically opposite angles
$a + b + c = 360^\circ$	angles at a point

“IT LOOKS LIKE IT” IS NOT A REASON

A diagram is a guide, not evidence: angles are rarely drawn to scale. The reason must be one of the four facts, named.

02

Worked examples

EXAMPLE 1 Two angles on a straight line are 125° and x . Find x .

1

$x + 125 = 180$
Angles on a line.

2

$x = 55^\circ$

ANSWER

55°

EXAMPLE 2 Angles of x and $2x$ lie on a straight line. Find both.

① $x + 2x = 180$

② $3x = 180$, so $x = 60$.

③ The angles are 60° and 120° .

Check: they add to 180° ✓

ANSWER

60° and 120°

EXAMPLE 3 Two crossing lines make angles of $3x$ and $x + 40$ vertically opposite. Find them.

① Vertically opposite angles are equal.

② $3x = x + 40$

③ $2x = 40$, so $x = 20$.

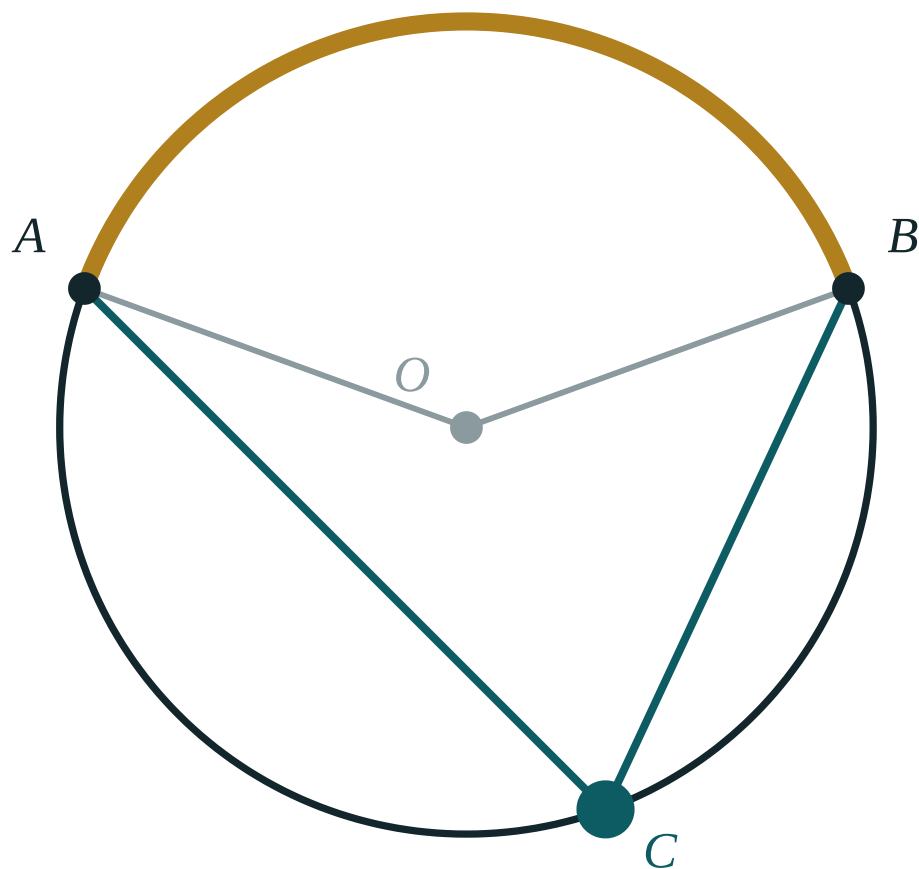
④ Each angle is 60° .

ANSWER

60°

03 Interactive model

Insist on a reason beside every line for a fortnight; the habit carries into Grade 7 proofs, where it becomes most of the mark.

Angles at a point add to 360° central $\angle AOB$ **140°** inscribed $\angle ACB$ **70°** ratio **2**arc AB **140°** **04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Angles on a line	Yoyiq burchak	Смежные углы
Angles at a point	Nuqtadagi burchaklar	Углы вокруг точки
Vertically opposite	Vertikal burchaklar	Вертикальные углы
Complementary	Yig'indisi 90°	Дополнительные до 90°
Supplementary	Yig'indisi 180°	Смежные
To justify	Asoslash	Обосновать
Unknown angle	Noma'lum burchak	Неизвестный угол
Reason	Sabab	Обоснование

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Angles on a straight line add to:

Angles at a point add to:

Vertically opposite angles are:

125° on a line with x gives:

45°

55°

65°

235°

x and $2x$ on a line give x equal to:

45

60

90

120

A geometry answer needs:

the number

the number and a reason

a diagram

a measurement

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 125° on a line with x

ANSWER 55°

2. 90° on a line with x

ANSWER 90°

3. x vertically opposite 68°

ANSWER 68°

4. x and 37° make a right angle

ANSWER 53°

5. 140° , 90° and x at a point

ANSWER 130°

6. Angles on a line add to

ANSWER 180°

7. Angles at a point add to

ANSWER 360°

Medium · the standard question

7 PROBLEMS

1. x and $2x$ on a line

ANSWER 60° and 120°

2. x , $2x$, $3x$ on a line

ANSWER 30° , 60° , 90°

3. $3x$ and $x + 40$ vertically opposite

ANSWER $x = 20$, each 60°

4. x , x , x , x at a point

ANSWER 90° each

5. $2x$ and $x + 30$ on a line: find x

ANSWER $x = 50$

6. Four angles at a point, three of them 80° , 100° , 60°

ANSWER 120°

7. Two supplementary angles differ by 40°

ANSWER 110° and 70°

Hard · stretch and reason

7 PROBLEMS

1. x , $x + 20$, $2x$ and 90° at a point

ANSWER $x = 62.5^\circ$

2. Two complementary angles in the ratio 2 : 3

ANSWER 36° and 54°

3. Two supplementary angles in the ratio 4 : 5

ANSWER 80° and 100°

4. An angle is 3 times its supplement

ANSWER 135° and 45°

5. An angle equals its own complement

ANSWER 45°

6. An angle is 20° more than twice its complement

ANSWER $66\frac{2}{3}^\circ$

7. Why is “it looks equal” not a reason?

ANSWER Diagrams are rarely drawn to scale

07 Homework

Homework — 5 tasks

Write a reason beside every line of working.

- Find x if x and 118° lie on a straight line.
- Find x if x , $3x$ and 140° meet at a point.
- Find x if $5x$ and $2x + 60$ are vertically opposite.
- Two supplementary angles are in the ratio 3 : 7. Find both.

5. Find the angle that is twice its own complement.